

Combining Instance Filtering and Attribute Weighting in Decision Tree and Naïve Bayes Hybrids

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Abstract—Naïve Bayes (NB) can remove training rows before a decision tree is fitted, while a tree can select and weight attributes for NB. We combine these steps as Algorithm 3 and test both orders with NB or C4.5 as the final classifier. Ten datasets are evaluated using ten repetitions of stratified 10-fold cross-validation, with cleaning fitted within each training fold and every held-out row scored. Mean accuracy is 81.21% and 80.89% for the combined tree methods, versus 86.33% for plain C4.5. The combined NB methods reach 75.63% and 76.35%, versus 75.93% for plain NB and 77.35% for attribute weighting alone. None of eight paired accuracy tests is significant after Holm adjustment. Cleaning before cross-validation raises the filter-tree score by 13.50 percentage points, but also removes cases from evaluation. Class-specific records show uneven deletion. These results do not establish a benefit from combining the steps.

Index Terms—Naïve Bayes, decision tree, instance filtering, attribute weighting, cross-validation

I. INTRODUCTION AND RELATED WORK

Farid et al. [1] proposed two hybrids. Algorithm 1 fits NB, removes training rows it misclassifies, and fits a tree on the retained rows. Algorithm 2 fits a tree, keeps the attributes it tests, and gives more weight to attributes used nearer the root when fitting NB. We combine these steps as Algorithm 3 and compare both orders. Removing rows first changes the sample used to select attributes; selecting attributes first changes the NB judge used to remove rows. Each order ends in either NB or C4.5 so that cleaning can be compared within each classifier.

Tree-derived NB weights were studied by Hall [2]. Brodley and Friedl [3] studied mislabeled data using several classifiers. Here, disagreement with one NB judge does not prove that a label is wrong. We use “filtering” to mean rejection by that judge, without treating rejected rows as verified label errors.

Before-CV filtering can use labels from future test folds and remove those cases from evaluation. Training-fold cleaning leaves all held-out cases for scoring. For filtering, the two procedures change both the information available and the evaluated sample. Cawley and Talbot [4] discuss evaluation bias; our comparison examines these procedures for the NB-

tree methods. We report accuracy, macro-F1, paired dataset-level tests, and class-specific deletion rates.

II. ALGORITHMS

A. Algorithm 1: instance filtering

Given training data D , the NB judge is fitted on D and predicts the labels of those same rows. A row remains only when the prediction agrees with its supplied label. Algorithm 1 then fits C4.5 on the retained rows. The reference arm ending in NB uses the same filter but refits NB as its final classifier. If deletion would leave fewer than two classes, our implementation retains the original training rows.

This is a training-set disagreement rule. Its predictions are not out-of-fold estimates of label quality. A judge can reject a correctly labeled case because the class distributions overlap or because its own model assumptions are unsuitable.

B. Algorithm 2: attribute selection and weighting

A C4.5 tree is fitted on the training data. Let d_j be the smallest depth at which it tests attribute j , with the root at depth one. The weight is

$$w_j = \begin{cases} 1/\sqrt{d_j}, & j \text{ is tested,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Untested attributes are dropped. The weighted NB class score is

$$s_c(x) = \log P(c) + \sum_{j \in S} w_j \log P(x_j | c), \quad (2)$$

where S is the retained attribute set. NB predicts the class with the largest score. The reference arm ending in a tree uses the selected columns, but its final C4.5 does not use NB’s attribute weights.

C. Algorithm 3: the combined hybrid

The instances-first path, denoted NA, runs Algorithm 1’s plain NB filter and then derives the selected attributes and weights from a tree on the retained rows. The attributes-first path, AN, derives the attributes and weights from the original

Algorithm 3 Combined cleaning and classification**Require:** Training data D , order $o \in \{\text{NA}, \text{AN}\}$, final learner $L \in \{\text{NB}, \text{DT}\}$

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1: if  $o = \text{NA}$  then
2:    $D \leftarrow \text{FILTER}(D, \text{plain NB})$ 
3:    $(S, W) \leftarrow \text{SELECT}(D, \text{C4.5})$ 
4: else
5:    $(S, W) \leftarrow \text{SELECT}(D, \text{C4.5})$ 
6:    $D \leftarrow D[:, S]$ 
7:    $D \leftarrow \text{FILTER}(D, \text{NB with } W)$ 
8: end if
9: if  $o = \text{NA}$  then
10:   $D \leftarrow D[:, S]$ 
11: end if
12: Fit  $M$  on  $D$  using  $L$  (use  $W$  if  $L = \text{NB}$ )
13: return  $M, S$ 

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TABLE I
DATASET SIZES IN THE FILES USED FOR EVALUATION

Dataset	Rows	Attributes	Classes
Breast cancer	286	9	2
Contact lenses	24	4	3
Diabetes	768	8	2
Glass	214	9	6
Iris	150	4	3
Soybean	683	35	19
Vote	435	16	2
Image segmentation	2310	19	7
Tic-tac-toe	958	9	2
NSL-KDD	25192	41	22

training fold, then uses weighted NB to filter its rows. In AN, the same weights reach both the judge and the final NB; they are not re-estimated after deletion. Each path ends in either weighted NB or C4.5.

Algorithm 3 uses the filtering safeguard stated above. If a selection tree tests no attribute, the combined path keeps all current columns with unit weights. No empty-selection event occurs in the stored runs. At prediction time, the test data keep the selected columns S and all their rows. Every held-out case must still be classified.

III. EXPERIMENTAL METHOD

A. Data and implementation

Table I lists our copies of the ten datasets named in the 2014 study. Image segmentation combines two UCI files; NSL-KDD uses its 20% training file [5]. Glass and NSL-KDD each contain one fewer class than the original study lists. These files allow comparisons on shared data but do not establish an exact reproduction of the original data split.

C4.5 is Weka J48 3.8.6 with pruning and defaults $-C \ 0.25 -M \ 2$ [6]. NB uses add-one nominal counts and Gaussian numeric likelihoods. Nominal attributes remain single columns, so tree selection refers to original attributes, not one-hot indicators. Category vocabularies come from the complete files and remain fixed across folds; missing nominal values form

a separate category. Counts, numeric distributions, and priors are fitted on training rows.

B. Protocols and methods

Protocol B is the main evaluation: stratified 10-fold CV with seeds 0–9, giving 100 folds per dataset and method. All methods share each split. The judge, selection tree, and final classifier are fitted on training rows. Only selected columns transfer to the test fold; no test row is removed. Learner settings are fixed, without nested hyperparameter search.

Protocol A cleans the whole dataset once, then cross-validates the final classifier on the cleaned data. Cleaning sees future test labels, and filtering changes the sample used to form folds. A and B folds are therefore not paired. Their score differences are descriptive and cannot separate label leakage from sample selection.

We compare plain NB and C4.5, Algorithm 1–DT, Algorithm 2–NB, and the four Algorithm 3 variants. Two reference methods, Algorithm 1–NB and Algorithm 2–DT, separate cleaning from the choice of final learner. Tables use N for filtering, A for attribute selection and weighting, and NA/AN for the two orders.

C. Measures and statistical tests

Each dataset’s accuracy and macro-F1 are averaged over 100 folds with equal fold weight; the summary gives each dataset equal weight. Macro-F1 averages class F1 over classes present in the fold’s true labels or predictions, using zero for undefined F1. Scores are not pooled across datasets. Table II reports means and standard deviations of the ten seed-level benchmark averages.

Following dataset-level comparisons [7], we use two-sided Wilcoxon signed-rank tests on ten dataset accuracy averages. The eight pairs are the original hybrids versus their baselines, four combined methods versus their baselines, and two order comparisons. Exact permutation enumeration includes zero differences; Holm adjustment covers all eight tests at 0.05. This test family was added during revision from saved results, not registered in advance. Differences use percentage points (pp). Single-step versus combined comparisons are descriptive; nonsignificance does not establish equivalence. For each filter, a class’s removal rate uses its training-row count as the denominator.

IV. RESULTS AND DISCUSSION

A. Combined methods have lower mean tree accuracy

Table II gives the benchmark averages, and Table III shows the individual datasets. Uncleaned C4.5 reaches 86.33% accuracy and 81.92% macro-F1. The two combined tree methods reach 81.21% and 80.89% accuracy, losses of 5.12 and 5.44 pp. Their macro-F1 scores also fall, to 75.76% and 73.93%. Adding attribute selection after filtering recovers only 0.18 pp relative to Algorithm 1’s 81.03% average accuracy.

For NB, attribute weighting alone gives the highest mean in this comparison, 77.35%, against 75.93% for plain NB. The combined methods reach 75.63% and 76.35%. Their

TABLE II
PROTOCOL B SCORES (%): MEAN $\pm$ STANDARD DEVIATION OF TEN
SEED-LEVEL BENCHMARK AVERAGES; DATASETS HAVE EQUAL WEIGHT

Method	Accuracy (%)	Macro-F1 (%)
NB	75.93 $\pm$ 0.32	69.80 $\pm$ 0.52
C4.5	86.33 $\pm$ 0.33	81.92 $\pm$ 0.44
N-NB	75.42 $\pm$ 0.46	69.08 $\pm$ 0.46
N-DT	81.03 $\pm$ 0.37	75.70 $\pm$ 0.63
A-NB	77.35 $\pm$ 0.41	71.73 $\pm$ 0.63
A-DT	86.19 $\pm$ 0.33	81.84 $\pm$ 0.42
NA-NB	75.63 $\pm$ 0.45	69.60 $\pm$ 0.72
NA-DT	81.21 $\pm$ 0.29	75.76 $\pm$ 0.58
AN-NB	76.35 $\pm$ 0.34	69.91 $\pm$ 0.60
AN-DT	80.89 $\pm$ 0.45	73.93 $\pm$ 0.80

changes relative to plain NB are -0.30 and $+0.42$ pp, and neither improves on Algorithm 2’s average. The reference arms show where scores change: filtering followed by NB reaches 75.42%, while selecting attributes before C4.5 leaves its mean almost unchanged at 86.19%.

Scores differ across datasets. On Glass, instances-first NB falls from the plain NB baseline of 45.95% to 30.93%; instances-first C4.5 falls from 66.58% to 48.97%. On Contact lenses, the same NB combination improves from 73.00% to 83.00%, although that dataset has only 24 rows. The 24-row result gives limited evidence of benefit.

Table IV reports the paired accuracy comparisons. AN-DT is lower than the C4.5 baseline with raw $p = 0.014$, but its Holm-adjusted $p = 0.109$ exceeds the stated threshold. None of the eight comparisons is significant after adjustment. These mean losses do not establish how the methods perform on other datasets.

Changing from NA to AN raises mean NB accuracy by 0.71 pp and lowers mean tree accuracy by 0.32 pp. The corresponding raw p values are 0.275 and 0.922. We find no clear order advantage under these settings. This finding is limited to a single pass through the two steps, with fixed learner settings and the weights passed to the AN judge.

B. Cleaning before the split changes the score

Table V compares the two evaluation protocols. Algorithm 1 reaches 94.52% after full-data filtering, against 81.03% when the judge is refitted within each training fold. The score increase is 13.50 pp. Algorithm 2 changes much less, from 77.35% to 78.07%, or 0.72 pp. The combined tree methods also rise sharply under Protocol A, to 94.72% and 96.32%.

Protocol B asks the final classifier to predict all held-out cases. Protocol A first uses the NB judge to choose which cases may reach the evaluation. The increase may reflect removal of difficult cases rather than improved classification on the full sample. Attribute-only cleaning retains all rows, although its full-data selection still uses information from future test folds. Its smaller change does not justify selecting attributes before CV.

C. Deletion changes class representation

The breast-cancer records show how uneven the removal can be. Across Protocol B folds, the plain NB judge removes an average of 50.63% of recurrence-event rows and 14.51% of no-recurrence-event rows. After attribute selection, the weighted judge removes 58.97% and 11.72%, respectively. These are mean class-specific removal rates, not percentages of the entire dataset.

The minority class loses a much larger share of its training examples in both paths. The records cannot establish whether those labels were wrong or whether NB failed on difficult but valid cases. They do show why a cleaner’s effect needs to be assessed through both classification scores and the training cases it discards.

V. LIMITATIONS

The benchmark consists of the original study’s dataset family, with the file differences described above. It does not establish performance on modern datasets or on data with a known amount of label noise. In particular, the NSL-KDD C4.5 baseline is 99.47%, compared with 71.11% reported in 2014. That discrepancy remains unresolved. The current comparison uses shared data and splits, but should not be described as an exact replication of the earlier scores.

Rare classes limit the evaluation. Contact lenses has classes with four and five rows, and NSL-KDD contains classes with fewer than ten examples, including singleton classes. Some stratified folds therefore lack a class in training or testing. Repeating the splits does not supply new examples, and the fold-specific macro-F1 convention can average different class sets. These results cannot support reliable rare-class performance claims.

Although model fitting and supervised cleaning are confined to training rows in Protocol B, nominal category vocabularies come from the complete files. The estimate is conditional on those known vocabularies; it does not test deployment with unseen nominal values. No calibration or out-of-fold judging is used to validate the filter’s rejection decisions. The tests involve ten datasets and can miss small effects, especially after multiplicity adjustment. Finally, the Protocol A comparison cannot identify which evaluation procedure the 2014 study used.

VI. CONCLUSION

Algorithm 3 combines NB instance filtering and tree-based attribute weighting in either order. On ten datasets under fold-specific evaluation, both combined tree methods have lower mean accuracy and macro-F1 than uncleaned C4.5, while both combined NB methods trail attribute weighting alone. The paired tests provide no clear order advantage. The before-CV scores and class-specific removal rates support a practical recommendation: fit supervised cleaning within the training fold, score every held-out row, and report which classes lose training cases.

TABLE III

PROTOCOL B ACCURACY BY DATASET (%). BASE IS THE UNCLEANNED CLASSIFIER; N IS INSTANCE FILTERING; A IS ATTRIBUTE SELECTION AND WEIGHTING; NA AND AN ARE THE TWO ALGORITHM 3 ORDERS. N-NB AND A-DT ARE REFERENCE ARMS.

Dataset	Final NB					Final C4.5				
	Base	N	A	NA	AN	Base	N	A	NA	AN
Breast cancer	72.12	72.68	72.43	71.28	72.53	74.84	71.77	73.91	71.77	72.08
Contact lenses	73.00	73.00	80.67	83.00	79.33	84.67	84.67	84.67	84.67	80.67
Diabetes	75.52	74.41	76.73	75.58	75.67	74.12	75.12	74.18	75.23	75.57
Glass	45.95	37.00	47.84	30.93	37.28	66.58	48.83	66.58	48.97	49.15
Segmentation	79.76	74.92	82.30	74.53	77.32	96.81	89.24	96.83	89.26	90.88
Iris	95.53	95.33	95.40	95.33	95.87	94.27	94.93	94.27	94.93	93.80
NSL-KDD	67.42	75.79	62.80	74.53	75.23	99.47	87.43	99.47	89.68	87.07
Soybean	90.09	89.00	89.40	85.67	86.66	92.71	88.71	92.26	87.89	90.98
Tic-tac-toe	69.83	71.89	70.68	70.03	68.07	84.74	74.43	84.74	74.55	73.17
Vote	90.06	90.13	95.24	95.45	95.49	95.08	95.13	95.03	95.13	95.54

TABLE IV

DATASET-LEVEL PAIRED ACCURACY TESTS. DIFFERENCES ARE FIRST METHOD MINUS SECOND; HOLM ADJUSTMENT COVERS ALL EIGHT COMPARISONS.

Comparison	Δ (pp)	p	Holm p
N-DT vs. DT	-5.30	0.055	0.383
A-NB vs. NB	+1.42	0.131	0.654
NA-DT vs. DT	-5.12	0.055	0.383
AN-DT vs. DT	-5.44	0.014	0.109
NA-NB vs. NB	-0.30	1.000	1.000
AN-NB vs. NB	+0.42	0.846	1.000
AN-DT vs. NA-DT	-0.32	0.922	1.000
AN-NB vs. NA-NB	+0.71	0.275	1.000

TABLE V

MEAN ACCURACY (%) UNDER THE TWO PROTOCOLS. BEFORE-CV FILTERING CHANGES THE EVALUATION SAMPLE; ITS SCORES ARE DIAGNOSTIC COMPARISONS.

Method	B (%)	A (%)	A-B (pp)
N-NB	75.42	91.93	+16.51
N-DT	81.03	94.52	+13.50
A-NB	77.35	78.07	+0.72
A-DT	86.19	86.52	+0.33
NA-NB	75.63	90.11	+14.48
NA-DT	81.21	94.72	+13.51
AN-NB	76.35	91.08	+14.74
AN-DT	80.89	96.32	+15.43

REFERENCES

- [1] D. M. Farid, L. Zhang, C. M. Rahman, M. A. Hossain, and R. Strachan, "Hybrid decision tree and naïve Bayes classifiers for multi-class classification tasks," *Expert Systems with Applications*, vol. 41, no. 4, pp. 1937–1946, 2014.
- [2] M. Hall, "A decision tree-based attribute weighting filter for naïve Bayes," *Knowledge-Based Systems*, vol. 20, no. 2, pp. 120–126, 2007.
- [3] C. E. Brodley and M. A. Friedl, "Identifying mislabeled training data," *Journal of Artificial Intelligence Research*, vol. 11, pp. 131–167, 1999.
- [4] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, no. 70, pp. 2079–2107, 2010.
- [5] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*. IEEE, 2009, pp. 1–6.
- [6] I. H. Witten, E. Frank, M. A. Hall, and C. J. Pal, *Data Mining: Practical*

Machine Learning Tools and Techniques, 4th ed. Cambridge, MA: Morgan Kaufmann, 2016.

- [7] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, no. 1, pp. 1–30, 2006.